Synchronizing Task Behavior: Aligning Multiple Tasks during Test-Time Training

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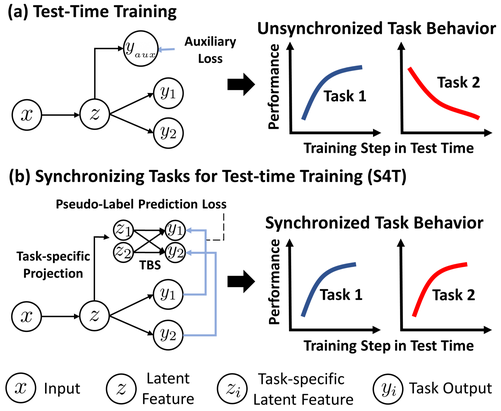
ICCV OCT 19-23, 2025 HONOLULU HAWAII

Background

- **Test-Time Training (TTT)** is a technique that allows a model to improve its performance during inference by using unlabeled test inputs.
- Generally, It adds a self-supervised task (like predicting image rotation or reconstructing masked features) and updates the model parameters on each test sample.

Motivation

- **Unsynchronized task adaptation.** Existing TTT methods treat each task independently, leading to misaligned adaptation behavior when applied to multi-task settings.
- **Limitations of auxiliary tasks.** A single auxiliary task often fails to generalize across different main tasks, causing inconsistent improvements in multi-task scenarios.
- **Need for task relationship modeling.** Incorporating task relations learned from the source domain can help synchronize adaptation across tasks and improve overall multi-task performance.



Main Method

- **Assumption 1: Preservation of Task Relation.** Inter-task relations learned in the source domain are structurally preserved in the target domain despite distribution shifts.
- **Assumption 2: Sufficient Training of TBS (Task Behavior Synchronizer).** The Task Behavior Synchronizer (TBS) can accurately predict task labels from masked task-specific latent features when it is sufficiently trained with an appropriate masking ratio.

Assumption 1 (Preservation of Task Relation) *The structural dependencies between latent variables and task labels remain consistent across source and target domains:*

$$p(y_{s,i}|z_{s,1},\ldots,z_{s,n}) \approx f(p(y_{t,i}|z_{t,1},\ldots,z_{t,n}))$$

for all $i \in \{1,2,\ldots,n\}$, where f represents a domain-dependent transformation.

Assumption 2 (Sufficient Training of TBS) *If the Task Behavior Synchronizer is sufficiently trained, it can reliably generate task labels from the masked latent space:*

$$p(y_i|z_1,\ldots,z_n) \approx p(y_i|\tilde{z}_1,\ldots,\tilde{z}_n)$$

for all $i \in \{1,2,\ldots,n\}$.

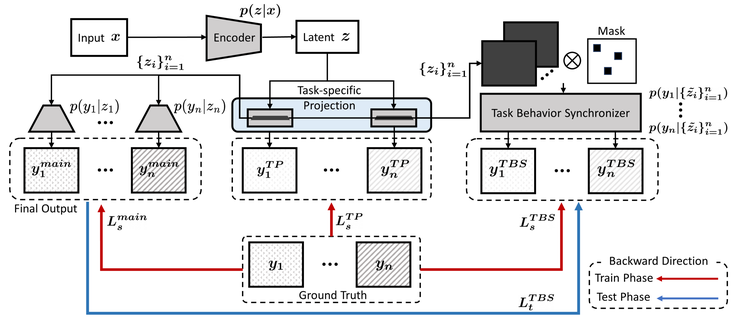
- **Proposition 1:** Target task loss is upper-bounded by masked prediction loss and its discrepancy from full latent predictions, justifying the use of task relations for adaptation.

Proposition 1 *Under Assumptions 1 and 2, the following inequality holds:*

$$d(\theta, p(\{z_{t,i}\}_{i=1}^n, y_{t,j})) \le d(\theta, p(\{\tilde{z}_{t,i}\}_{i=1}^n, y_{t,j})) \quad (1)$$
$$+ \mathbb{E}_{p(\{z_{t,i}\}_{i=1}^n, \{\tilde{z}_{t,i}\}_{i=1}^n)}[D]. \quad (2)$$

where $D = d\left[p_\theta(y_{t,j}|\{z_{t,i}\}_{i=1}^n), p_\theta(y_{t,j}|\{\tilde{z}_{t,i}\}_{i=1}^n)\right]$.

- At test time, the input image is encoded and projected into task-specific latent features.
- The main decoder produces each task's output, while a separate **Task Behavior Synchronizer (TBS)** predicts task labels from masked task-specific latents.
- By minimizing the difference between TBS predictions and main outputs, the model adapts using learned task relations, achieving synchronized multi-task adaptation under domain shift.

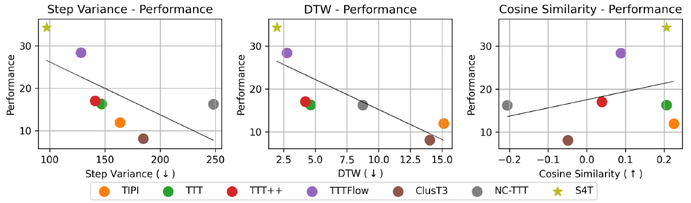


Experimental Results

- S4T consistently outperforms prior methods by achieving higher accuracy and better task synchronization in diverse domain shift scenarios.

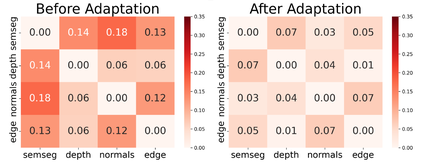
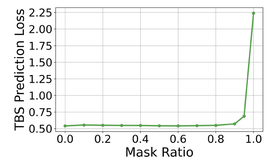
Dataset	Taskonomy → NYUD-v2					Taskonomy → PASCAL-Context			
Task	Semseg	Depth	Normal	Edge	$\triangle_{TTT}$	Semseg	Normal	Edge	$\triangle_{TTT}$
Metric	mIoU ↑	RMSE ↓	mErr ↓	RMSE ↓	% ↑	mIoU ↑	mErr ↓	RMSE ↓	% ↑
Base	29.31 ±6.30e-2	1.179 ±8.00e-3	61.32 ±8.20e-1	0.1443 ±7.10e-5	+0.00	27.08 ±1.40e-2	63.46 ±9.50e-1	0.1185 ±7.10e-5	0.00
TENT [49]	40.42 ±1.09e+0	1.056 ±1.70e-2	56.09 ±3.21e+0	0.1441 ±2.10e-4	+14.26	40.65 ±1.30e-1	58.76 ±2.05e+0	0.1183 ±9.00e-5	+19.26
TIPI [36]	48.12 ±1.78e+0	1.029 ±6.50e-2	55.71 ±4.93e-1	0.1440 ±6.40e-5	+21.57	43.01 ±1.30e-5	39.03 ±2.30e-4	0.1186 ±1.40e-8	+32.41
TTT [46]	41.31 ±4.46e-1	1.061 ±1.00e-3	47.54 ±4.31e-1	0.1440 ±2.83e-5	+18.43	39.29 ±2.30e-1	33.76 ±9.04e-1	0.1183 ±4.00e-5	+30.70
TTT++ [29]	43.97 ±1.11e+0	1.107 ±1.50e-2	46.71 ±6.40e-2	0.1440 ±2.60e-5	+20.05	37.26 ±5.00e-2	36.87 ±4.50e-2	0.1183 ±2.40e-4	+26.55
TTTFlow [38]	52.75 ±7.50e-2	1.075 ±1.00e-3	46.02 ±1.25e-1	0.1442 ±4.70e-5	+28.47	38.73 ±2.45e-1	43.30 ±7.55e-1	0.1184 ±4.70e-5	+27.97
ClusT3 [8]	41.88 ±2.06e+0	1.102 ±1.40e-2	46.52 ±7.00e-3	0.1440 ±7.30e-6	+18.44	33.26 ±1.20e-5	35.31 ±3.70e-5	0.1184 ±3.60e-8	+22.43
ActMAD [32]	27.62 ±1.61e-1	1.193 ±8.00e-3	56.53 ±2.31e+0	0.1444 ±3.00e-5	+0.20	22.09 ±1.00e-5	54.73 ±1.00e-5	0.1188 ±1.90e-8	-1.63
NC-TTT [39]	48.17 ±7.00e+0	1.086 ±5.20e-2	48.32 ±1.23e+0	0.1440 ±2.80e-5	+23.40	42.81 ±1.10e-1	40.65 ±1.10e-1	0.1184 ±7.70e-6	+31.37
S4T (ours)	59.37 ±1.52e-1	1.052 ±7.50e-3	45.33 ±7.20e-2	0.1441 ±5.10e-5	**+34.94**	45.42 ±1.90e-1	41.41 ±6.30e-1	0.1183 ±5.00e-5	**+34.20**

- We assess multi-task synchronization using Step Variance (SV), Dynamic Time Warping (DTW) and Cosine Similarity.
- S4T shows **synchronized adaptation** across tasks under domain shift.



- Reliability of Assumptions 1: Inter-task relations are preserved after domain shift.
- Reliability of Assumptions 2: Task-Behavior Synchronizer can infer labels from masked latents.

Task affinity gap before and after adaptation



TBS prediction loss with varying masking ratios